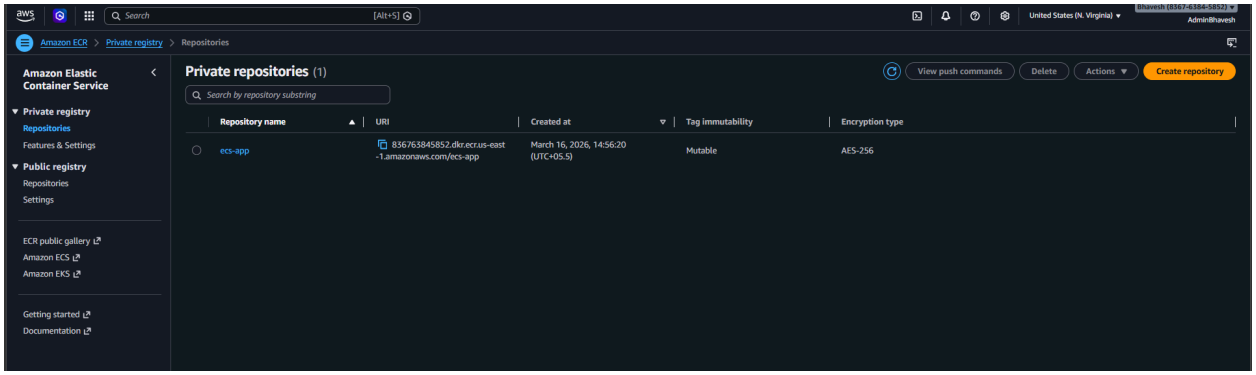
AKIA4FUYDATOHZUJMQ7X

KwfCC2hcz1hvHNAVH1XLxNxd3yYymK/IfibW3o+0

ECR



App

app.py

```
from flask import Flask, render_template, jsonify, request
import pymysql
import os

app = Flask(__name__)

DB_HOST = os.getenv("DB_HOST")
DB_USER = os.getenv("DB_USER")
DB_PASS = os.getenv("DB_PASS")
DB_NAME = os.getenv("DB_NAME")

def get_connection():
    return pymysql.connect(
        host=DB_HOST,
        user=DB_USER,
        password=DB_PASS,
        database=DB_NAME
    )
```

```
@app.route("/")
def home():
    return render_template("index.html")

@app.route("/api/users", methods=["GET"])
def get_users():
    conn = get_connection()
    cursor = conn.cursor()

    cursor.execute("SELECT name FROM users")
    rows = cursor.fetchall()

    users = [r[0] for r in rows]

    conn.close()

    return jsonify(users)

@app.route("/api/users", methods=["POST"])
def add_user():

    data = request.json
    name = data["name"]

    conn = get_connection()
    cursor = conn.cursor()

    cursor.execute("INSERT INTO users(name) VALUES(%s)", (name,))
    conn.commit()

    conn.close()

    return jsonify({"message": "User added"})


if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

index.html

```html
<!DOCTYPE html>
<html>
<head>
<title>AWS Container App</title>
</head>

<body>

<h1>AWS DevOps Demo App</h1>

<input id="username" placeholder="Enter name">
<button onclick="addUser()">Add User</button>

<h2>Users</h2>
<ul id="userList"></ul>

<script src="/static/script.js"></script>

</body>
</html>
```

script.js

```javascript
async function loadUsers() {

    let res = await fetch("/api/users");
    let users = await res.json();

    let list = document.getElementById("userList");
    list.innerHTML = "";

    users.forEach(u => {
        let li = document.createElement("li");
        li.innerText = u;
        list.appendChild(li);
    });

}
```

```
async function addUser(){

    let name = document.getElementById("username").value;

    await fetch("/api/users",{
        method:"POST",
        headers:{
            "Content-Type":"application/json"
        },
        body:JSON.stringify({name:name})
    });

    loadUsers();
}

loadUsers();
```

requirements.txt

```
flask
pymysql
```

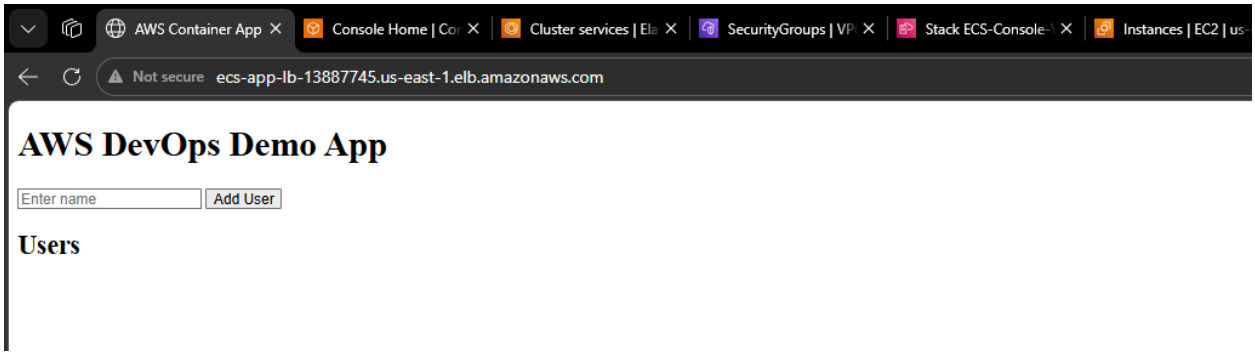
dockerfile

```
FROM python:3.10

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

EXPOSE 5000

CMD ["python","app.py"]
```

Created Registry

```
[ec2-user@ip-10-0-0-231 aws-container-app]$ aws ecr create-repository --repository-name ecs-app --region us-east-1
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:us-east-1:836763845852:repository/ecs-app",
        "registryId": "836763845852",
        "repositoryName": "ecs-app",
        "repositoryUri": "836763845852.dkr.ecr.us-east-1.amazonaws.com/ecs-app",
        "createdAt": "2026-03-16T09:26:20.659000+00:00",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
```

Pushed to Registry

```
[ec2-user@ip-10-0-0-231 aws-container-app]$ docker tag app:latest 836763845852.dkr.ecr.us-east-1.amazonaws.com/ecs-app:l
atest
[ec2-user@ip-10-0-0-231 aws-container-app]$ docker images
REPOSITORY                                          TAG       IMAGE ID       CREATED          SIZE
836763845852.dkr.ecr.ap-south-1.amazonaws.com/app   latest    b7f9881b9f19   24 minutes ago   1.11GB
836763845852.dkr.ecr.us-east-1.amazonaws.com/app    latest    b7f9881b9f19   24 minutes ago   1.11GB
836763845852.dkr.ecr.us-east-1.amazonaws.com/ecs-app latest   b7f9881b9f19   24 minutes ago   1.11GB
app                                                 latest    b7f9881b9f19   24 minutes ago   1.11GB
[ec2-user@ip-10-0-0-231 aws-container-app]$ docker push 836763845852.dkr.ecr.us-east-1.amazonaws.com/ecs-app:latest
The push refers to repository [836763845852.dkr.ecr.us-east-1.amazonaws.com/ecs-app]
4d625b85d46a: Pushed
f4017cec4f21: Pushed
2d124cc1a1f0: Pushed
151ef4fa4100: Pushed
92567dd46c62: Pushed
2cc4db6b38b2: Pushed
84d9300b8400: Pushed
481dc13bd3b0: Pushed
7fe06297fe28: Pushed
a541a5ccaa30: Pushed
latest: digest: sha256:290e762c549081ad144fa8e0b6d7ab776f1da49afe1769fb91a880e8d1d3395e size: 2421
[ec2-user@ip-10-0-0-231 aws-container-app]$
```

Output